



Graph and Coordinate by **Divyanshu**

DAY 07

## Graph and Coordinate

### — The Language of Position —

Every map, every scatter plot, every image an AI "sees" is built on one idea: the **coordinate system**. Today we cover coordinate geometry up to Class 10 level and connect it directly to how AI systems use coordinates, distances and graphs.

**37+**

SOLVED QUESTIONS

**7**

CORE CONCEPTS

**Class 6-10**

LEVEL COVERED

**AI**

CONNECT BOXES

# Graph and Coordinate

Welcome back to **AI School**! Whenever an AI system plots your location on a map, places a data point on a scatter graph, or reads the pixels of an image, it is using **coordinates**. Today we build coordinate geometry from the ground up — plotting points, measuring distances, dividing line segments, finding areas, and understanding slope — and connect each idea directly to how AI "sees" and reasons using graphs.

## Why This Matters for AI

Every image is a grid of coordinates (row, column) — that's literally how a computer stores a photo. Every dataset used to train an AI model is a cloud of points in "feature space", found and compared using the exact same distance and slope formulas you'll practise today. Coordinate geometry is the visual backbone of Artificial Intelligence.

## Today's Learning Goals

- ✓ Understand the Cartesian plane, axes, origin and the four quadrants.
- ✓ Plot points and find reflections across the axes.
- ✓ Apply the Distance Formula between two points.
- ✓ Use the Section & Midpoint Formula to divide a line segment.
- ✓ Find the area of a triangle from its vertices, and test collinearity.
- ✓ Find the slope of a line and form its equation; connect this to AI graphs.

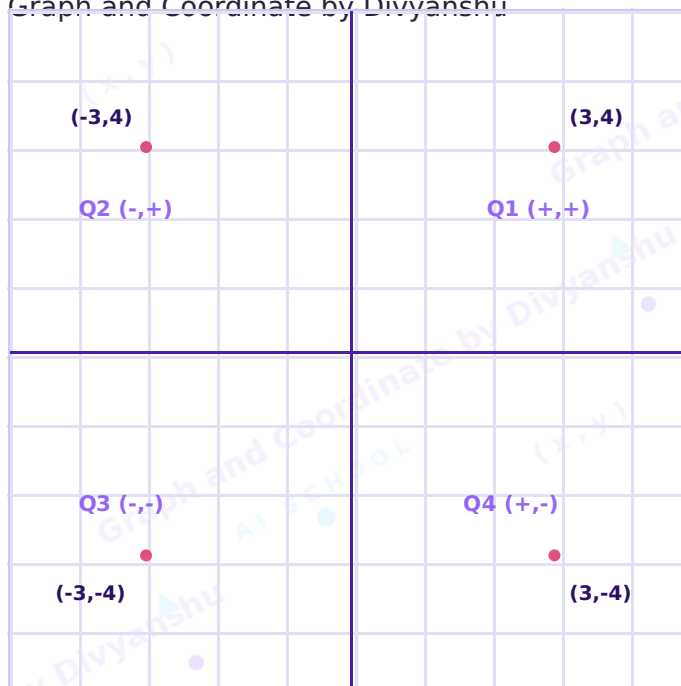
Class 6-8: Cartesian Plane & Plotting

Class 9: Coordinate Geometry Basics

Class 10: Distance, Section, Area & Lines

## 1 The Cartesian Plane & Quadrants

A **Cartesian plane** is formed by two perpendicular number lines: the horizontal **x-axis** and the vertical **y-axis**. They cross at the **origin O(0, 0)**. Every point on the plane is written as an **ordered pair (x, y)**, where x is the distance along the x-axis and y is the distance along the y-axis.



| Quadrant | Sign of (x, y) | Example  |
|----------|----------------|----------|
| I        | (+, +)         | (3, 4)   |
| II       | (-, +)         | (-3, 4)  |
| III      | (-, -)         | (-3, -4) |
| IV       | (+, -)         | (3, -4)  |

A point with  $y = 0$  lies **on the x-axis**. A point with  $x = 0$  lies **on the y-axis**. The origin is  $(0, 0)$ .



### AI Connect

A digital photo is a grid of pixels addressed by coordinates (row, column) — exactly a Cartesian plane! When an AI model "crops" or "detects an object" in an image, it is really just working with rectangles of  $(x, y)$  coordinates.

## Solved Examples — Set A: Quadrants & Axes

### Example 1

EASY

In which quadrant does the point  $(3, 4)$  lie?

**Solution:**

Both  $x = 3$  and  $y = 4$  are positive.

**Quadrant I**

### Example 2

EASY

In which quadrant does the point  $(-2, 5)$  lie?

**Solution:**

$x = -2$  (negative),  $y = 5$  (positive).

**Quadrant II**

### Example 3

EASY

In which quadrant does the point  $(-4, -6)$  lie?

**Solution:**

Both  $x = -4$  and  $y = -6$  are negative.

**Quadrant III**

### Example 4

EASY

In which quadrant does the point  $(5, -3)$  lie?

**Solution:**

$x = 5$  (positive),  $y = -3$  (negative).

**Quadrant IV**

### Example 5

MEDIUM

State the position of the points  $(7, 0)$  and  $(0, -3)$ .

**Solution:**

$(7, 0)$  has  $y = 0$ , so it lies on the x-axis.

$(0, -3)$  has  $x = 0$ , so it lies on the y-axis.

**$(7, 0) \rightarrow$  x-axis,  $(0, -3) \rightarrow$  y-axis**

## 2 Plotting Points & Reflections

To plot a point  $(x, y)$ : start at the origin, move  $x$  units right/left (right if  $x$  is positive, left if negative), then move  $y$  units up/down (up if  $y$  is positive, down if negative).

### Reflection Rules

- Reflection across the **x-axis**:  $(x, y) \rightarrow (x, -y)$
- Reflection across the **y-axis**:  $(x, y) \rightarrow (-x, y)$
- Reflection through the **origin**:  $(x, y) \rightarrow (-x, -y)$

### AI Connect

Flipping an image horizontally or vertically — a common trick used to create more training data for AI models ("data augmentation") — is nothing but reflecting every pixel's coordinates across an axis, exactly like the rules above!

## Solved Examples — Set B: Plotting & Reflections

### Example 6

**EASY**

Find the reflection of the point  $(3, 5)$  across the x-axis.

**Solution:**

Rule:  $(x, y) \rightarrow (x, -y)$

**Reflection =  $(3, -5)$**

### Example 7

**EASY**

Find the reflection of the point  $(3, 5)$  across the y-axis.

**Solution:**

Rule:  $(x, y) \rightarrow (-x, y)$

**Reflection =  $(-3, 5)$**

### Example 8

**MEDIUM**

A point is 4 units to the right and 3 units above the origin. Write its coordinates, then find its reflection through the origin.

**Solution:**

Coordinates of the point =  $(4, 3)$

Reflection through origin:  $(x, y) \rightarrow (-x, -y) = (-4, -3)$

**Point =  $(4, 3)$ , Reflection =  $(-4, -3)$**

### 3 The Distance Formula

The distance between two points  $A(x_1, y_1)$  and  $B(x_2, y_2)$  is found using a formula built directly from the Pythagoras Theorem:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



#### AI Connect: Nearest Neighbours

A very popular AI algorithm called **K-Nearest Neighbours (KNN)** classifies a new data point by literally computing the distance formula between it and every known point, then picking the closest ones. Every time an app says "customers similar to you also liked...", the distance formula is running behind the scenes.

### Solved Examples — Set C: Distance Formula

#### Example 9

EASY

Find the distance between  $A(2, 3)$  and  $B(5, 7)$ .

**Solution:**

$$d = \sqrt{(5-2)^2 + (7-3)^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25}$$

$$d = 5 \text{ units}$$

#### Example 10

MEDIUM

Find the distance between  $A(-1, 2)$  and  $B(3, -1)$ .

**Solution:**

$$d = \sqrt{(3-(-1))^2 + (-1-2)^2} = \sqrt{4^2 + (-3)^2} = \sqrt{16+9} = \sqrt{25}$$

$$d = 5 \text{ units}$$

#### Example 11

EASY

Find the distance between  $A(1, 1)$  and  $B(4, 5)$ .

**Solution:**

$$d = \sqrt{(4-1)^2 + (5-1)^2} = \sqrt{9+16} = \sqrt{25}$$

$$d = 5 \text{ units}$$

**Example 12****HARD**

Show that the triangle with vertices  $O(0, 0)$ ,  $A(3, 0)$  and  $B(0, 4)$  is a right-angled triangle.

**Solution:**

$$OA = \sqrt{[(3-0)^2 + (0-0)^2]} = \sqrt{9} = 3$$

$$OB = \sqrt{[(0-0)^2 + (4-0)^2]} = \sqrt{16} = 4$$

$$AB = \sqrt{[(0-3)^2 + (4-0)^2]} = \sqrt{9+16} = \sqrt{25} = 5$$

$$\text{Check Pythagoras: } OA^2 + OB^2 = 9 + 16 = 25 = AB^2 \checkmark$$

**Right-angled at O, since  $OA^2 + OB^2 = AB^2$**

**Example 13 — AI Connect****MEDIUM**

In a recommendation system, User A is plotted at  $(2, 3)$  and User B at  $(6, 6)$  based on their movie-rating habits. Find the "similarity distance" between them.

**Solution:**

$$d = \sqrt{[(6-2)^2 + (6-3)^2]} = \sqrt{16+9} = \sqrt{25}$$

**Distance = 5 units (smaller distance = more similar users)**

## 4 Section Formula & Midpoint

The **Section Formula** gives the coordinates of a point P that divides the line segment joining A(x1, y1) and B(x2, y2) internally in the ratio m : n:

$$P = \left( \frac{m \cdot x_2 + n \cdot x_1}{m+n}, \frac{m \cdot y_2 + n \cdot y_1}{m+n} \right)$$

When P is exactly halfway (ratio 1:1), this becomes the simpler **Midpoint Formula**:

$$\text{Midpoint} = \left( \frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$$



### AI Connect

When an AI-based image editor "morphs" one face into another, or when an animation smoothly moves an object between two positions on screen, it is calculating in-between points using exactly the section formula, frame by frame.

## Solved Examples — Set D: Section & Midpoint Formula

### Example 14

**MEDIUM**

Find the point that divides the segment joining A(1, 2) and B(7, 8) in the ratio 1 : 2.

**Solution:**

$$x = \frac{1 \times 7 + 2 \times 1}{1+2} = \frac{7+2}{3} = \frac{9}{3} = 3$$

$$y = \frac{1 \times 8 + 2 \times 2}{1+2} = \frac{8+4}{3} = \frac{12}{3} = 4$$

$$P = (3, 4)$$

### Example 15

**EASY**

Find the midpoint of A(1, 2) and B(7, 8).

**Solution:**

$$\text{Midpoint} = \left( \frac{1+7}{2}, \frac{2+8}{2} \right) = \left( \frac{8}{2}, \frac{10}{2} \right)$$

$$\text{Midpoint} = (4, 5)$$

**Example 16****HARD**

Find the point dividing the segment joining A(2, 2) and B(12, 12) in the ratio 2 : 3.

**Solution:**

$$x = (2 \times 12 + 3 \times 2) / (2 + 3) = (24 + 6) / 5 = 30 / 5 = 6$$

$$y = (2 \times 12 + 3 \times 2) / (2 + 3) = 30 / 5 = 6 \text{ (same, since } x \text{ and } y \text{ values match on this segment)}$$

$$\mathbf{P = (6, 6)}$$

## 5 Area of a Triangle & Collinearity

Given three vertices  $A(x_1, y_1)$ ,  $B(x_2, y_2)$ ,  $C(x_3, y_3)$ , the area of triangle ABC is:

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

### Collinearity Check

If three points lie on the same straight line ("collinear"), the "triangle" they form has **zero area**. So, to check if points are collinear, just calculate the area — if it comes out 0, the points are collinear!

### AI Connect

Computer vision systems use this exact area formula to check whether three detected points (say, on a road lane or a robot's path) fall on a straight line, or to compute the area of a detected shape on a satellite image.

## Solved Examples — Set E: Area & Collinearity

### Example 17

MEDIUM

Find the area of the triangle with vertices  $(1, 1)$ ,  $(4, 1)$ ,  $(4, 5)$ .

**Solution:**

$$\begin{aligned} \text{Area} &= \frac{1}{2} |1(1-5) + 4(5-1) + 4(1-1)| \\ &= \frac{1}{2} |1(-4) + 4(4) + 4(0)| = \frac{1}{2} |-4+16+0| = \frac{1}{2} (12) \end{aligned}$$

**Area = 6 sq. units**

### Example 18

MEDIUM

Find the area of the triangle with vertices  $(0, 0)$ ,  $(4, 0)$ ,  $(0, 3)$ .

**Solution:**

$$\text{Area} = \frac{1}{2} |0(0-3) + 4(3-0) + 0(0-0)| = \frac{1}{2} |0+12+0|$$

**Area = 6 sq. units**

**Example 19****HARD**

Check whether the points (1, 2), (3, 6) and (5, 10) are collinear.

**Solution:**

$$\begin{aligned}\text{Area} &= \frac{1}{2} |1(6-10) + 3(10-2) + 5(2-6)| \\ &= \frac{1}{2} |1(-4) + 3(8) + 5(-4)| = \frac{1}{2} |-4+24-20| = \frac{1}{2} (0)\end{aligned}$$

**Area = 0, so the points ARE collinear**

## 6 Slope of a Line & Equation of a Line

The **slope (m)** of a line through A(x1, y1) and B(x2, y2) measures its steepness:

$$m = (y_2 - y_1) / (x_2 - x_1)$$

Once we know the slope and one point, we can write the equation of the line using the **point-slope form**:  $y - y_1 = m(x - x_1)$ , which simplifies to the familiar  **$y = mx + c$**  form (m = slope, c = y-intercept).



### AI Connect: The Line That Powers Machine Learning

The simplest AI prediction model, **linear regression**, does exactly one thing: it finds the "best-fit" straight line  $y = mx + c$  through a cloud of data points. The **slope** tells the AI how strongly the input affects the output, and training the model is really just an automatic, large-scale way of calculating m and c!

## Solved Examples — Set F: Slope & Line Equations

### Example 20

EASY

Find the slope of the line through (1, 2) and (3, 8).

**Solution:**

$$m = (8-2)/(3-1) = 6/2$$

$$m = 3$$

### Example 21

MEDIUM

Find the slope of the line through (2, 5) and (2, 9).

**Solution:**

$$m = (9-5)/(2-2) = 4/0 \rightarrow \text{division by zero}$$

**Slope is undefined — the line is vertical**

### Example 22

MEDIUM

Find the equation of the line passing through (1, 2) with slope 3.

**Solution:**

$$y - y_1 = m(x - x_1) \rightarrow y - 2 = 3(x - 1)$$

$$y - 2 = 3x - 3 \rightarrow y = 3x - 1$$

$$\text{Equation: } y = 3x - 1$$

**Example 23****HARD**

Find the equation of the line passing through (0, 1) and (2, 5).

**Solution:**

$$\text{Slope } m = (5-1)/(2-0) = 4/2 = 2$$

$$\text{Using } (0,1): y - 1 = 2(x - 0) \rightarrow y = 2x + 1$$

$$\text{Equation: } y = 2x + 1$$

**Example 24****MEDIUM**

Draw up a table of values for the line  $y = 2x + 1$  using  $x = -1, 0, 1, 2$ .

**Solution:**

$$x = -1 \rightarrow y = 2(-1)+1 = -1$$

$$x = 0 \rightarrow y = 2(0)+1 = 1$$

$$x = 1 \rightarrow y = 2(1)+1 = 3$$

$$x = 2 \rightarrow y = 2(2)+1 = 5$$

**Points:  $(-1,-1), (0,1), (1,3), (2,5)$  — all lie on one straight line**

## 7 AI Corner — Graphs Inside Real AI Systems

Coordinate geometry doesn't stay inside the textbook — it is the visual language AI engineers use every single day. Here are two direct applications, fully solved:

### Example 25 — Nearest Movie Recommendation

MEDIUM

A user's taste is plotted at point  $U(4, 4)$ . Movie 1 is at  $(2, 3)$  and Movie 2 is at  $(7, 7)$ . Using the distance formula, which movie should the AI recommend first?

**Solution:**

$$\text{Distance to Movie 1} = \sqrt{(4-2)^2 + (4-3)^2} = \sqrt{4+1} = \sqrt{5} \approx 2.24$$

$$\text{Distance to Movie 2} = \sqrt{(4-7)^2 + (4-7)^2} = \sqrt{9+9} = \sqrt{18} \approx 4.24$$

**Movie 1 is closer (smaller distance) → recommend Movie 1 first**

### Example 26 — Watching an AI "Learn" on a Graph

HARD

While training, an AI model's error ("loss") is plotted against its weight  $w$ : at  $w=1$ ,  $\text{loss}=9$ ; at  $w=2$ ,  $\text{loss}=4$ ; at  $w=3$ ,  $\text{loss}=1$ . Find the slope between the first two points and between the last two points to see how fast the loss is dropping.

**Solution:**

$$\text{Slope from } (1,9) \text{ to } (2,4): m = (4-9)/(2-1) = -5/1 = -5$$

$$\text{Slope from } (2,4) \text{ to } (3,1): m = (1-4)/(3-2) = -3/1 = -3$$

**Both slopes are negative (loss is decreasing); the AI is learning and improving as  $w$  increases**

### 💡 Big Idea

Whether it's comparing users, tracking a robot's path, or watching a model "train" — an AI system is constantly plotting points, measuring distances and reading slopes. Coordinate geometry is the shared visual language between mathematics and machines.

## ✓ Quick Recap

1. The **quadrants**, split by the x-axis and y-axis, meeting at the origin (0,0).  
4
2. Reflections flip the sign of x, y, or both, depending on the axis.
3. **Distance formula** :  $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$  — the basis of AI "nearest neighbour" search.
4. **Section/ midpoint formula** finds a point that divides a segment in a given ratio.
5. **Area formula** from coordinates also tells us if 3 points are collinear (area = 0).
6. **Slope (m)** measures steepness;  $y = mx + c$  is exactly the formula behind linear regression in AI.

## 📖 Formula Sheet

|                           |                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------|
| <b>Distance Formula</b>   | $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$                                           |
| <b>Section Formula</b>    | $P = ((m \cdot x_2 + n \cdot x_1) / (m + n), (m \cdot y_2 + n \cdot y_1) / (m + n))$ |
| <b>Midpoint Formula</b>   | $M = ((x_1 + x_2) / 2, (y_1 + y_2) / 2)$                                             |
| <b>Area of Triangle</b>   | $\frac{1}{2}  x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) $                     |
| <b>Slope of a Line</b>    | $m = (y_2 - y_1) / (x_2 - x_1)$                                                      |
| <b>Equation of a Line</b> | $y - y_1 = m(x - x_1), \text{ or } y = mx + c$                                       |

**8 Extra Practice Zone (Fully Solved)**

More solved questions mixing every idea from today — try covering the solution and solving first, then check your work!

**Practice 1****EASY**

Which quadrant does  $(-7, -2)$  lie in?

**Solution:**

Both coordinates are negative.

**Quadrant III**

**Practice 2****EASY**

Find the distance between  $(3, 3)$  and  $(7, 6)$ .

**Solution:**

$$d = \sqrt{[(7-3)^2 + (6-3)^2]} = \sqrt{[16+9]} = \sqrt{25}$$

**$d = 5$  units**

**Practice 3****EASY**

Find the midpoint of  $(-2, -3)$  and  $(6, 7)$ .

**Solution:**

$$\text{Midpoint} = ((-2+6)/2, (-3+7)/2) = (4/2, 4/2)$$

**Midpoint =  $(2, 2)$**

**Practice 4****MEDIUM**

Find the point dividing the segment joining  $(0, 0)$  and  $(10, 10)$  in the ratio  $3 : 2$ .

**Solution:**

$$x = (3 \times 10 + 2 \times 0)/(3+2) = 30/5 = 6$$

$$y = (3 \times 10 + 2 \times 0)/(3+2) = 30/5 = 6$$

**$P = (6, 6)$**

**Practice 5****MEDIUM**

Find the area of the triangle with vertices (2, 3), (6, 3), (6, 7).

**Solution:**

$$\begin{aligned}\text{Area} &= \frac{1}{2} |2(3-7) + 6(7-3) + 6(3-3)| \\ &= \frac{1}{2} |2(-4) + 6(4) + 6(0)| = \frac{1}{2} |-8+24+0| = \frac{1}{2} (16)\end{aligned}$$

**Area = 8 sq. units**

**Practice 6****EASY**

Find the slope of the line joining (0, 0) and (4, 4).

**Solution:**

$$m = (4-0)/(4-0) = 4/4$$

**m = 1**

**Practice 7****MEDIUM**

Find the equation of the line with slope -2 passing through (1, 1).

**Solution:**

$$y - 1 = -2(x - 1) \rightarrow y - 1 = -2x + 2$$

**y = -2x + 3**

**Practice 8****HARD**

Check whether (1, 1), (2, 2) and (3, 3) are collinear.

**Solution:**

$$\begin{aligned}\text{Area} &= \frac{1}{2} |1(2-3) + 2(3-1) + 3(1-2)| \\ &= \frac{1}{2} |1(-1) + 2(2) + 3(-1)| = \frac{1}{2} |-1+4-3| = \frac{1}{2} (0)\end{aligned}$$

**Area = 0 → the points ARE collinear**

**Practice 9****EASY**

Reflect the point (-5, 2) across the y-axis.

**Solution:**

$$\text{Rule: } (x, y) \rightarrow (-x, y)$$

**Reflection = (5, 2)**

### Practice 10 — AI Connect

MEDIUM

A data point in an AI model's "feature space" is at (3, 4), representing (brightness, contrast) of an image. Find its distance from the origin — this is called the "magnitude" of the data vector.

**Solution:**

$$d = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{9+16} = \sqrt{25}$$

**Magnitude = 5 units**

## ★ Fun Facts & Self-Check

### ✨ Did You Know?

- The Cartesian plane is named after French mathematician **René Descartes**, who reportedly got the idea while watching a fly move across the corner of his ceiling — realising its position could always be described by two numbers!
- Google Maps, GPS navigation, and every video game character's on-screen position are all powered by coordinate systems just like the one you studied today.
- In AI, a "feature space" can have hundreds or even thousands of coordinates (dimensions) instead of just  $x$  and  $y$  — but the same distance and slope ideas still apply.

### ✓ Self-Check Before You Move to Day 8

- ☐ I can identify the quadrant of any point and plot points on a Cartesian plane.
- ☐ I can find the reflection of a point across the  $x$ -axis,  $y$ -axis, or origin.
- ☐ I can apply the distance formula between any two points.
- ☐ I can use the section and midpoint formulas to divide a line segment.
- ☐ I can find the area of a triangle from coordinates and test collinearity.
- ☐ I can find the slope of a line, form its equation, and connect it to AI's "best-fit line".

### 🔭 Coming Up: Day 8

Next session we explore **graphing linear equations in two variables** in more depth and see how AI models visualise decision boundaries on a graph — the next step toward understanding how machines "draw the line" between categories.

AI School · Foundations Track · Day 7 Worksheet · Keep practising, keep plotting! 🚀